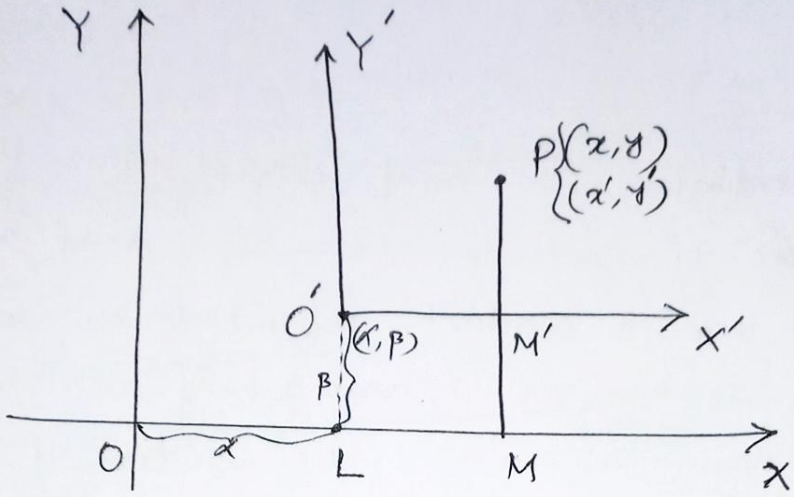
(Transformation of coordinates with respect to change of origin or axes:)

Transformation of coordinates :

The coordinates of a point depend upon the origin and the axes of coordinates chosen. If a point be given in position, then its coordinates referred to one set of axes will change as soon as a new set of axes be taken. That is, the above coordinates of the equation of the curve changes when the origin is changed or the direction of axes changed or both. It is evident that the equation of a curve will also change by such a transformation. But if we know the equation of the curve referred to one set of axes, we can deduce the equation referred to another set of axes. When we pass from one set of axes to another, the process is known as the transformation of coordinates. In other words, the process of changing the coordinates of a point or the equation of a curve is called transformation of coordinates.

To change the origin of coordinates without changing the direction of axes (Translation of axes):

Let, OX and OY be the original axes of coordinates. Let us take a new pair of axes O'X' and O'Y' parallel to the old pair OX and OY; Let, (α, β) be the coordinates of the new origin O' referred to the original axes.



Suppose, P be any point whose coordinates referred to the old axes are $P(x,y)$ and referred to the new axes $P(x',y')$. Draw PM perpendicular to OX, meeting $O'X'$ at M' and OX at M, respectively. From new origin O', draw $O'L$ perpendicular to OX. Then, we have,

$$OL = \alpha, \quad O'L = \beta, \quad OM = x, \quad PM = y$$

Also, $O'M' = x'$ & $PM' = y'$

From the geometry of the figure, we get,

$$OM = OL + LM = OL + O'M'$$

$$\Rightarrow x = \alpha + x'$$

Similarly,

$$PM = M'M + PM' = O'L + PM'$$

$$\Rightarrow y = \beta + y'$$

Therefore, the transformed coordinates are

$$\left.\begin{aligned} x &= x' + \alpha \\ y &= y' + \beta \end{aligned}\right\} \quad \text{——— (1) [old to new]}$$

$$\& \quad \left.\begin{aligned} x' &= x - \alpha \\ y' &= y - \beta \end{aligned}\right\} \quad \text{——— (2) [new to old]}$$

Here, (1) and (2) are the equations of transformation from the old to the new axes and from the new to the old, respectively.

Thus, when the origin is transferred to (α, β), the transformed equation of a curve is obtained by substituting $x'+\alpha$ for x and $y'+\beta$ for y and finally suppressing the accents. That is, the result is obtained only by putting $x+\alpha$ for x and $y+\beta$ for y in the original equation.

—o—

Example:

What does the equation $x^2+y^2-4x-6y+6=0$ become when the origin is transferred to $(2,3)$, the direction of the axes remaining unaltered?

Solution:

The given equation is

$$x^2 + y^2 - 4x - 6y + 6 = 0 \quad \text{——— (1)}$$

Putting $x = x' + 2$ and $y = y' + 3$ into equⁿ ① we have

$$(x'+2)^2 + (y'+3)^2 - 4(x'+2) - 6(y'+3) + 6 = 0$$

$$\Rightarrow x'^2 + 4x' + 4 + y'^2 + 6y' + 9 - 4x' - 8 - 6y' - 18 + 6 = 0$$

$$\Rightarrow x'^2 + y'^2 - 7 = 0$$

Removing suffixes we get

$$x^2 + y^2 - 7 = 0$$

Ans

Example:

Determine the equation of the curve $2x^2 + 3y^2 - 8x + 6y - 7 = 0$ when the origin is transferred to the point $(2, -1)$.

Solution:

The given equation is

$$2x^2 + 3y^2 - 8x + 6y - 7 = 0 \quad \text{——————} \quad ①$$

Putting $x = x' + 2$ and $y = y' - 1$ into equⁿ ① we obtain

$$2(x'+2)^2 + 3(y'-1)^2 - 8(x'+2) + 6(y'-1) - 7 = 0$$

$$\Rightarrow 2(x'^2 + 4x' + 4) + 3(y'^2 - 2y' + 1) - 8x' - 16 + 6y' - 6 - 7 = 0$$

$$\Rightarrow 2x'^2 + 8x' + 8 + 3y'^2 - 6y' + 3 - 8x' + 6y' - 29 = 0$$

$$\Rightarrow 2x'^2 + 3y'^2 = 18$$

Removing suffixes we have

$$2x^2 + 3y^2 = 18$$

(Ans)

Example:

Determine the equation of the curve $2x^2 + y^2 - 4x + 4y = 0$ when the origin is transferred to the point $(1, -2)$.

Solution:

The given equation is

$$2x^2 + y^2 - 4x + 4y = 0 \quad \text{——————} \quad (1)$$

Putting $x = x' + 1$ & $y = y' - 2$ into eqn (1) we obtain

$$2(x'+1)^2 + (y'-2)^2 - 4(x'+1) + 4(y'-2) = 0$$

$$\Rightarrow 2(x'^2 + 2x' + 1) + y'^2 - 4y' + 4 - 4x' - 4 + 4y' - 8 = 0$$

$$\Rightarrow 2x'^2 + 4x' + 2 + y'^2 - 4y' - 4x' + 4y' - 8 = 0$$

$$\Rightarrow 2x'^2 + y'^2 - 6 = 0$$

$$\Rightarrow 2x'^2 + y'^2 = 6$$

Removing suffixes we can write

$$2x^2 + y^2 = 6$$

(Ans)

Example:

Determine the equation of the curve $x^2-6x+2y^2+7=0$ when the origin is transferred to the point $(3,1)$.

Solution:

The given equation is

$$x^2-6x+2y^2+7=0 \quad \text{——————} (1)$$

Putting $x=x'+3$ & $y=y'+1$ into equ^n (1) we obt

$$(x'+3)^2-6(x'+3)+2(y'+1)^2+7=0$$

$$\Rightarrow x'^2+6x'+9-6x'-18+2y'^2+4y'+2+7=0$$

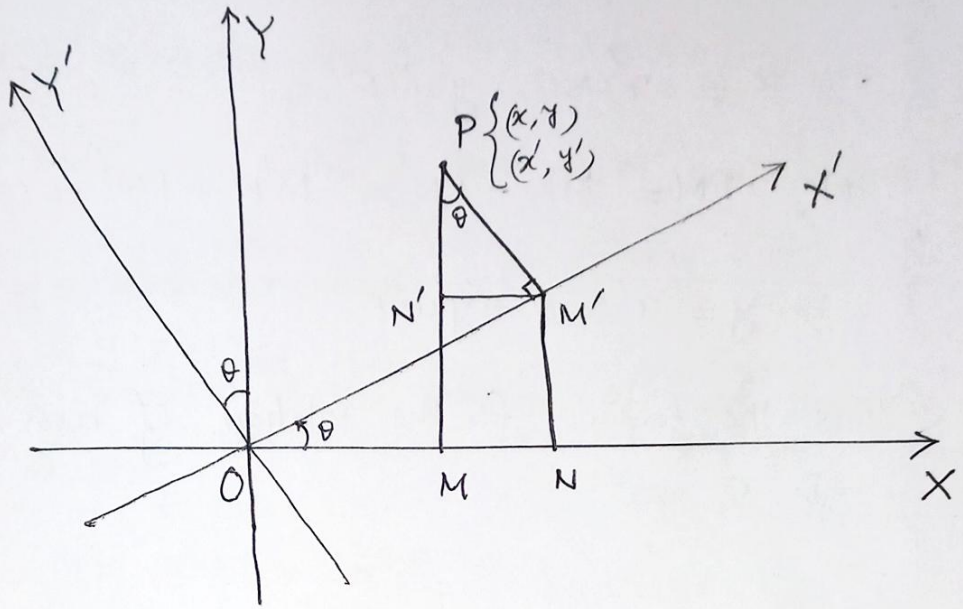
$$\Rightarrow x'^2+2y'^2+4y'=0$$

Removing suffixes we can write

$$x^2+2y^2+4y=0$$

Ans

To transform from one set of rectangular axes system to another with the same origin (Rotation of axes):



Let, OX and OY be the original axes of coordinates, and OX' and OY' be the new axes through the same origin and θ be the angle through which they have been turned in the same sense. That is $\angle XOX' = \angle YOY' = \theta$.

Let P be any point whose coordinates are (x, y) referred to the original axes and (x', y') referred to the new axes of coordinates. Draw PM and PM' perpendiculars to OX and OX', respectively. Draw M'N parallel to OY cutting OX at N and M'N' parallel to OX cutting PM at N'. Then we can write,

$OM = x$; $OM' = x'$; $PM = y$; $PM' = y'$;

Again, $PM \perp OX$ and $PM' \perp OX'$

$\therefore \angle MPM' = \angle XOX' = \theta$

Now, using the geometry of the figure, we have

$$OM = ON - MN = OM'\cos\theta - PM'\sin\theta$$

$$\Rightarrow x = x'\cos\theta - y'\sin\theta$$

And, $PM = MN' + PN' = M'N + PN' = OM'\sin\theta + PM'\cos\theta$

$$\Rightarrow y = x'\sin\theta + y'\cos\theta$$

Thus, the formula for the rotation of axes through an angle θ are:

$$\left.\begin{aligned} x &= x'\cos\theta - y'\sin\theta \\ y &= x'\sin\theta + y'\cos\theta \end{aligned}\right\} \quad \text{——— (1)}$$

Solving the equations in (1), we get

$$\left.\begin{aligned} x' &= x\cos\theta + y\sin\theta \\ y' &= -x\sin\theta + y\cos\theta \end{aligned}\right\} \quad \text{——— (2)}$$

These transforming equations (1) & (2) may be conveniently remembered from the following scheme:

	x'	y'
x	$\cos\theta$	$-\sin\theta$
y	$\sin\theta$	$\cos\theta$

which may be read either horizontally or vertically. The transformed equation of a curve, in this transformation, is then obtained by substituting $x\cos\theta - y\sin\theta$ for x and $x\sin\theta + y\cos\theta$ for y.

Change of origin with the change of the direction of axes:

If the origin is first transferred to the point (α, β) and the new set of axes through this origin be inclined at an angle θ to the original axes, the formulae of transformation will obviously be the combination of the above two sets of formulae, i.e.,

$$\left.\begin{aligned} x &= \alpha + x'\cos\theta - y'\sin\theta \\ \text{and } y &= \beta + x'\sin\theta + y'\cos\theta \end{aligned}\right\} \quad \text{——— (3)}$$

Example:

Transform to axes inclined at 45° to the original axes of the following equations:

(i) $x^2 - y^2 = a^2$

(ii) $x^2 - y^2 - 2\sqrt{2}x - 10\sqrt{2}y + 2 = 0$

Solution:

(i) Given that,

$$x^2 - y^2 = a^2 \quad \text{——— (1)}$$

When the axes have been rotated through an angle 45°, then we have,

$$x = x'\cos 45^\circ - y'\sin 45^\circ = \frac{1}{\sqrt{2}}(x' - y')$$

$$y = x'\sin 45^\circ + y'\cos 45^\circ = \frac{1}{\sqrt{2}}(x' + y')$$

Substituting the values of x & y into eqn (1), we get

$$\left\{\frac{1}{\sqrt{2}}(x' - y')\right\}^2 - \left\{\frac{1}{\sqrt{2}}(x' + y')\right\}^2 = a^2$$

$$\Rightarrow \frac{1}{2}(x'^2 - 2x'y' + y'^2) - \frac{1}{2}(x'^2 + 2x'y' + y'^2) = a^2$$

$$\Rightarrow \cancel{\frac{1}{2}x'^2} - x'y' + \cancel{\frac{1}{2}y'^2} - \cancel{\frac{1}{2}x'^2} - x'y' - \cancel{\frac{1}{2}y'^2} = a^2$$

$$\Rightarrow -2x'y' = a^2$$

$$\Rightarrow 2x'y' + a^2 = 0$$

Now dropping the suffixes, the equation becomes

$$2xy + a^2 = 0 \quad \text{——— (2)}$$

Hence, equⁿ (2) is the required transformed equation.

Ans

(ii) Given that,

$$x^2 - y^2 - 2\sqrt{2}x - 10\sqrt{2}y + 2 = 0 \quad \text{——— (1)}$$

When the axes have been rotated through an angle 45°, then we have,

$$x = x'\cos 45^\circ - y'\sin 45^\circ = \frac{1}{\sqrt{2}}(x' - y')$$

$$y = x'\sin 45^\circ + y'\cos 45^\circ = \frac{1}{\sqrt{2}}(x' + y')$$

Substituting the values of x and y into equⁿ (1), we get

$$\left\{\frac{1}{\sqrt{2}}(x'-y')\right\}^2 - \left\{\frac{1}{\sqrt{2}}(x'+y')\right\}^2 - 2\sqrt{2}\cdot\frac{1}{\sqrt{2}}(x'-y') - 10\sqrt{2}\cdot\frac{1}{\sqrt{2}}(x'+y') + 2 = 0$$

$$\Rightarrow \frac{1}{2}(x'^2 - 2x'y' + y'^2) - \frac{1}{2}(x'^2 + 2x'y' + y'^2) - 2x' + 2y' - 10x' - 10y' + 2 = 0$$

$$\Rightarrow \cancel{\frac{1}{2}x'^2} - x'y' + \cancel{\frac{1}{2}y'^2} - \cancel{\frac{1}{2}x'^2} - x'y' - \cancel{\frac{1}{2}y'^2} - 12x' - 8y' + 2 = 0$$

$$\Rightarrow -2x'y' - 12x' - 8y' + 2 = 0$$

$$\Rightarrow x'y' + 6x' + 4y' = 1$$

Now dropping the suffixes, the equation becomes,

$$xy + 6x + 4y = 1 \quad \text{——— (2)}$$

Thus, equⁿ (2) is the required transformed equation.

(Ans)

Example:

Suppose that the axes of an xy-coordinate system are rotated through an angle of $\theta = 45°$ to obtain an $x'y'$-coordinate system. Find the equation of the curve $x^2 - xy + y^2 - 6 = 0$ in $x'y'$-coordinates.

Solution:

Given that,

$$x^2 - xy + y^2 - 6 = 0 \quad \text{——— (1)}$$

If an xy-coordinate system is gone through a rotation of $45°$ of the axes, then we have

$$x = x'\cos 45° - y'\sin 45° = \frac{1}{\sqrt{2}}(x' - y')$$

& $$y = x'\sin 45° + y'\cos 45° = \frac{1}{\sqrt{2}}(x' + y')$$

Substituting the values of x & y into the given equⁿ (1) yields

$$\left\{\frac{1}{\sqrt{2}}(x' - y')\right\}^2 - \frac{1}{\sqrt{2}}(x' - y') \cdot \frac{1}{\sqrt{2}}(x' + y') + \left\{\frac{1}{\sqrt{2}}(x' + y')\right\}^2 - 6 = 0$$

$$\Rightarrow \frac{1}{2}(x'^2 - 2x'y' + y'^2) - \frac{1}{2}(x'^2 - y'^2) + \frac{1}{2}(x'^2 + 2x'y' + y'^2) - 6 = 0$$

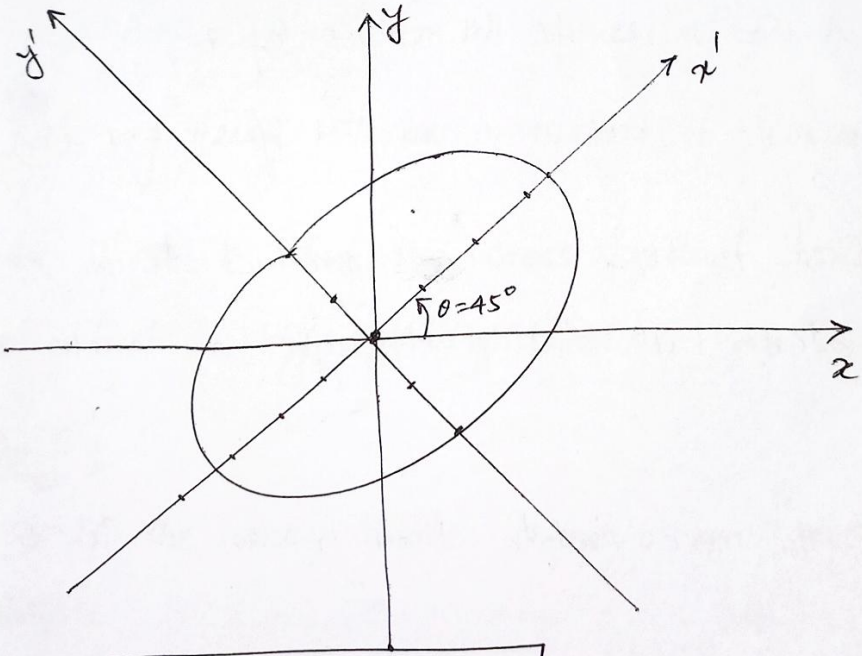
$$\Rightarrow \frac{1}{2}\left(x'^2 - 2x'y' + y'^2 - x'^2 + y'^2 + x'^2 + 2x'y' + y'^2\right) - 6 = 0$$

$$\Rightarrow \frac{1}{2}\left(x'^2 + 3y'^2\right) - 6 = 0$$

$$\Rightarrow x'^2 + 3y'^2 = 12$$

$$\Rightarrow \frac{x'^2}{12} + \frac{y'^2}{4} = 1$$

which is the equation of an ellipse shown in the following figure.



$$\boxed{x^2 - xy + y^2 - 6 = 0}$$

Theorem:

If the equation $Ax^2+Bxy+Cy^2+Dx+Ey+F=0$ ——— (1) is such that $B\neq 0$, and if an $x'y'$-coordinate system is obtained by rotating the xy-axes through an angle θ, then prove that,

(a) the expression (1) changes to

$$A'x'^2+B'x'y'+C'y'^2+D'x'+E'y'+F'=0$$

(b) $B^2-4AC=B'^2-4A'C'$ for all values of θ.

(c) $A+C=A'+C'$ for all values of θ.

(d) xy term will be vanished if $\cot 2\theta=\frac{A-C}{B}$

(e) if $A=C$ then the cross product term can be eliminated by rotating through an angle $\theta=45^\circ$.

Proof:

(a) If the axes rotate through an angle θ, then we have

$$x=x'\cos\theta-y'\sin\theta$$

& $$y=x'\sin\theta+y'\cos\theta$$

Substituting the values of x & y into equ^n (1) yields

$$A(x'\cos\theta-y'\sin\theta)^2+B(x'\cos\theta-y'\sin\theta)(x'\sin\theta+y'\cos\theta)$$
$$+C(x'\sin\theta+y'\cos\theta)^2+D(x'\cos\theta-y'\sin\theta)+E(x'\sin\theta+y'\cos\theta)+F$$

$$\Rightarrow A(x'^2\cos^2\theta - x'y'\sin 2\theta + y'^2\sin^2\theta)$$
$$+ B(x'^2\sin\theta\cos\theta - x'y'\sin^2\theta + x'y'\cos^2\theta - y'^2\sin\theta\cos\theta)$$
$$+ C(x'^2\sin^2\theta + x'y'\sin 2\theta + y'^2\cos^2\theta)$$
$$+ D(x'\cos\theta - y'\sin\theta) + E(x'\sin\theta + y'\cos\theta) + F = 0$$

$$\Rightarrow (A\cos^2\theta + B\cos\theta\sin\theta + C\sin^2\theta)x'^2$$
$$+ \{B(\cos^2\theta - \sin^2\theta) + (C - A)\sin 2\theta\}x'y'$$
$$+ (A\sin^2\theta - B\sin\theta\cos\theta + C\cos^2\theta)y'^2$$
$$+ (D\cos\theta + E\sin\theta)x' + (-D\sin\theta + E\cos\theta)y' + F = 0$$

$$\Rightarrow A'x'^2 + B'x'y' + C'y'^2 + D'x' + E'y' + F' = 0$$

where,

$$A' = A\cos^2\theta + B\cos\theta\sin\theta + C\sin^2\theta$$

$$B' = B\cos 2\theta + (C - A)\sin 2\theta$$

$$C' = A\sin^2\theta - B\sin\theta\cos\theta + C\cos^2\theta$$

$$D' = D\cos\theta + E\sin\theta$$

$$E' = -D\sin\theta + E\cos\theta$$

$$F' = F$$

(b) $B'^2 - 4A'C' = \{B\cos 2\theta + (C-A)\sin 2\theta\}^2$

$- 4(A\cos^2\theta + B\sin\theta\cos\theta + C\sin^2\theta)$.

$(A\sin^2\theta - B\sin\theta\cos\theta + C\cos^2\theta)$

$= \{B\cos 2\theta + (C-A)\sin 2\theta\}^2$

$- 2(A\cos^2\theta + B\sin\theta\cos\theta + C\sin^2\theta)$

$\cdot 2(A\sin^2\theta - B\sin\theta\cos\theta + C\cos^2\theta)$

$= \{B\cos 2\theta + (C-A)\sin 2\theta\}^2$

$- (A\cdot 2\cos^2\theta + B\cdot 2\sin\theta\cos\theta + C\cdot 2\sin^2\theta)$

$\cdot (A\,2\sin^2\theta - B\cdot 2\sin\theta\cos\theta + C\cdot 2\cos^2\theta)$

$= \{B\cos 2\theta + (C-A)\sin 2\theta\}^2$

$- (A(1+\cos 2\theta) + B\sin 2\theta + C(1-\cos 2\theta))$

$\cdot (A(1-\cos 2\theta) - B\sin 2\theta + C(1+\cos 2\theta))$

$= \{B\cos 2\theta + (C-A)\sin 2\theta\}^2$

$- \{(A+C) + (A-C)\cos 2\theta + B\sin 2\theta\}\{(A+C) - (A-C)\cos 2\theta - B\sin 2\theta\}$

$$= \left\{B\cos 2\theta + (C-A)\sin 2\theta\right\}^2$$

$$-\left[(A+C) + \left\{(A-C)\cos 2\theta + B\sin 2\theta\right\}\right]$$

$$\cdot\left[(A+C) - \left\{(A-C)\cos 2\theta + B\sin 2\theta\right\}\right]$$

$$= B^2\cos^2 2\theta + 2B(C-A)\sin 2\theta\cdot\cos 2\theta + (C-A)^2\sin^2 2\theta$$

$$-\left[(A+C)^2 - \left\{(A-C)\cos 2\theta + B\sin 2\theta\right\}^2\right]$$

$$= B^2\cos^2 2\theta + 2B(C-A)\sin 2\theta\cdot\cos 2\theta + (C-A)^2\sin^2 2\theta$$

$$-(A+C)^2 + \left\{(A-C)\cos 2\theta + B\sin 2\theta\right\}^2$$

$$= B^2\cos^2 2\theta + 2B(C-A)\sin 2\theta\cos 2\theta + (C-A)^2\sin^2 2\theta$$

$$-(A+C)^2 + (A-C)^2\cos^2 2\theta + 2B(A-C)\sin 2\theta\cos 2\theta$$

$$+ B^2\sin^2 2\theta$$

$$= B^2(\cos^2 2\theta + \sin^2 2\theta) - \cancel{2B(A-C)\sin 2\theta\cdot\cos 2\theta}$$

$$+(C-A)^2(\sin^2 2\theta + \cos^2 2\theta) - (A+C)^2$$

$$\cancel{+\,2B(A-C)\sin 2\theta\cos 2\theta}$$

$$= B^2 + (C-A)^2 - (C+A)^2$$

$$= B^2 - \left[(C+A)^2 - (C-A)^2\right]$$

$\therefore B'^2 - 4A'C' = B^2 - 4AC$

(Proved)

(c) $A' + C' = A\cos^2\theta + B\cos\theta\sin\theta + C\sin^2\theta$

$+ A\sin^2\theta - B\sin\theta\cos\theta + C\cos^2\theta$

$\Rightarrow A' + C' = A(\cos^2\theta + \sin^2\theta) + C(\sin^2\theta + \cos^2\theta)$

$\therefore A' + C' = A + C$

(Proved)

(d) xy term will be vanished if

$B' = 0$

$\Rightarrow B\cos 2\theta + (C - A)\sin 2\theta = 0$

$\Rightarrow B\cos 2\theta = (A - C)\sin 2\theta$

$\Rightarrow \cot 2\theta = \frac{A - C}{B}$

$\therefore \cot 2\theta = \frac{A - C}{B}$

(Showed)

(e) We have already proved that

$\cot 2\theta = \frac{A - C}{B}$ ——— (ii)

If $A = C$, then equⁿ (11) becomes

$$\cot 2\theta = \frac{0}{B} = 0 = \cot 90^\circ$$

$$\Rightarrow \quad 2\theta = 90^\circ$$

$$\therefore \quad \theta = 45^\circ$$

(Proved)

Example:

Identify and sketch the curve

$$153x^2 - 192xy + 97y^2 - 30x - 40y - 200 = 0$$

Solution:

Given that,

$$153x^2 - 192xy + 97y^2 - 30x - 40y - 200 = 0 \quad \text{——— (1)}$$

Here we have,

$$A = 153; \quad B = -192; \quad C = 97$$

$$\therefore \cot 2\theta = \frac{A - C}{B} = \frac{153 - 97}{-192} = -\frac{7}{24}$$

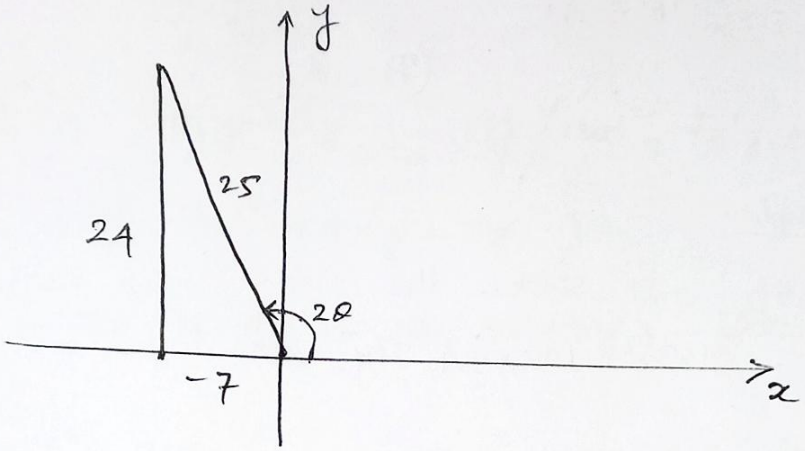
$$\Rightarrow \cot 2\theta = -7/24$$

Since θ is to be chosen in the range $0 < \theta < \pi/2$, this relationship is represented by the triangle shown in fig. 1. From that triangle we obtain $\cos 2\theta = -7/25$, which

implies that

$$\cos\theta = \sqrt{\frac{1+\cos 2\theta}{2}} = \sqrt{\frac{1-7/25}{2}} = 3/5$$

$$\sin\theta = \sqrt{\frac{1-\cos 2\theta}{2}} = \sqrt{\frac{1+7/25}{2}} = 4/5$$



We know that

$$\left.\begin{aligned} x &= x'\cos\theta - y'\sin\theta \\ y &= x'\sin\theta + y'\cos\theta \end{aligned}\right\} \longrightarrow (2)$$

Substituting the values of $\cos\theta$ & $\sin\theta$ into eqn (2), we have,

$$\left.\begin{aligned} x &= \frac{3}{5}x' - \frac{4}{5}y' \\ \& \quad y &= \frac{4}{5}x' + \frac{3}{5}y' \end{aligned}\right\} \longrightarrow (3)$$

Substituting the values of (3) into (1) yields

$$\frac{153}{25}(3x'-4y')^2 - \frac{192}{25}(3x'-4y')(4x'+3y') + \frac{97}{25}(4x'+3y')^2 - \frac{30}{5}(3x'-4y') - \frac{40}{5}(4x'+3y') - 200 = 0$$

$$\Rightarrow \frac{153}{25}(9x'^2 - 24x'y' + 16y'^2) - \frac{192}{25}(12x'^2 - 16x'y' + 9x'y' - 12y'^2)$$
$$+ \frac{97}{25}(16x'^2 + 24x'y' + 9y'^2)$$
$$- \frac{30}{5}(3x' - 4y') - \frac{40}{5}(4x' + 3y') - 200 = 0$$

$$\Rightarrow 153(9x'^2 - 24x'y' + 16y'^2) - 192(12x'^2 - 7x'y' - 12y'^2)$$
$$+ 97(16x'^2 + 24x'y' + 9y'^2)$$
$$- 150(3x' - 4y') - 200(4x' + 3y') - 5000 = 0$$

$$\Rightarrow 1377x'^2 - 3672x'y' + 2448y'^2 - 2304x'^2 + 1344x'y'$$
$$+ 2304y'^2 + 1552x'^2 + 2328x'y' + 873y'^2$$
$$- 450x' + 600y' - 800x' - 600y' - 5000 = 0$$

$$\Rightarrow 625x'^2 + 5625y'^2 - 1250x' - 5000 = 0$$

$$\Rightarrow 25x'^2 + 225y'^2 - 50x' - 200 = 0$$

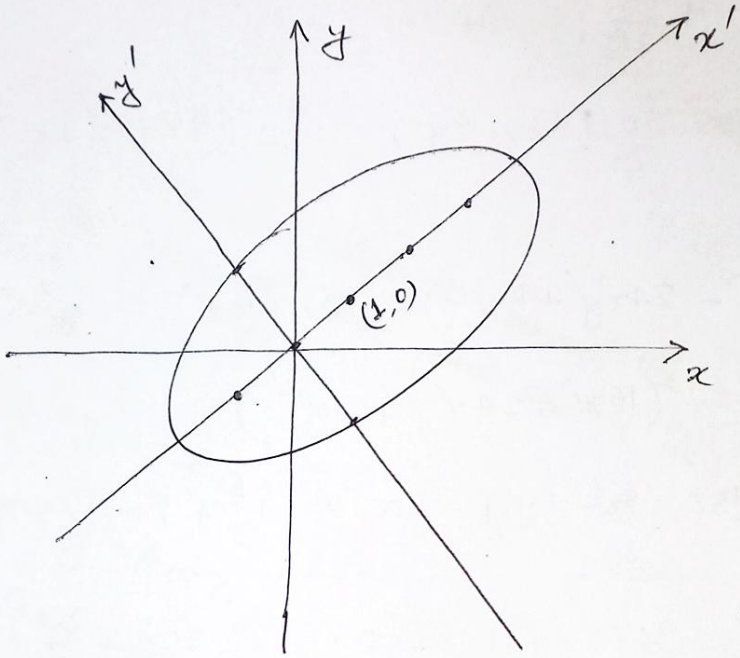
$$\Rightarrow x'^2 + 9y'^2 - 2x' - 8 = 0$$

$$\Rightarrow x'^2 - 2x' + 1 + 9y'^2 - 9 = 0$$

$$\Rightarrow (x' - 1)^2 + 9y'^2 = 9$$

$$\Rightarrow \frac{(x' - 1)^2}{9} + y'^2 = 1$$

$$\Rightarrow \frac{(x'-1)^2}{3^2} + \frac{y'^2}{1^2} = 1 \qquad \text{——— (4)}$$



Thus, equⁿ (4) is the equation in the $x'y'$– coordinate syste of an ellipse with centre $(1,0)$ in that coordinate system and the length of the semi major axis $a = 3$ and that of the semi minor axis $b = 1$.